

APPLIED ANATOMY-II-

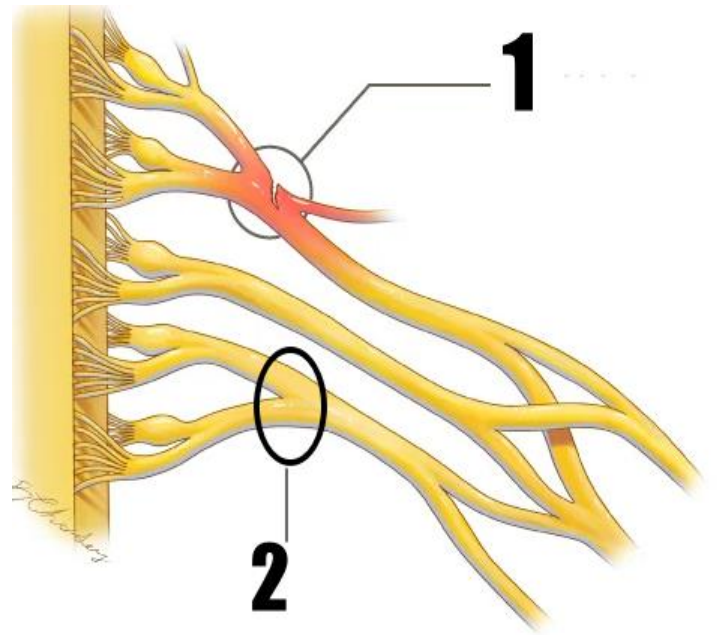
BRACHIAL PLEXUS

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Brachial Plexus Injuries

Key Anatomical Clues:

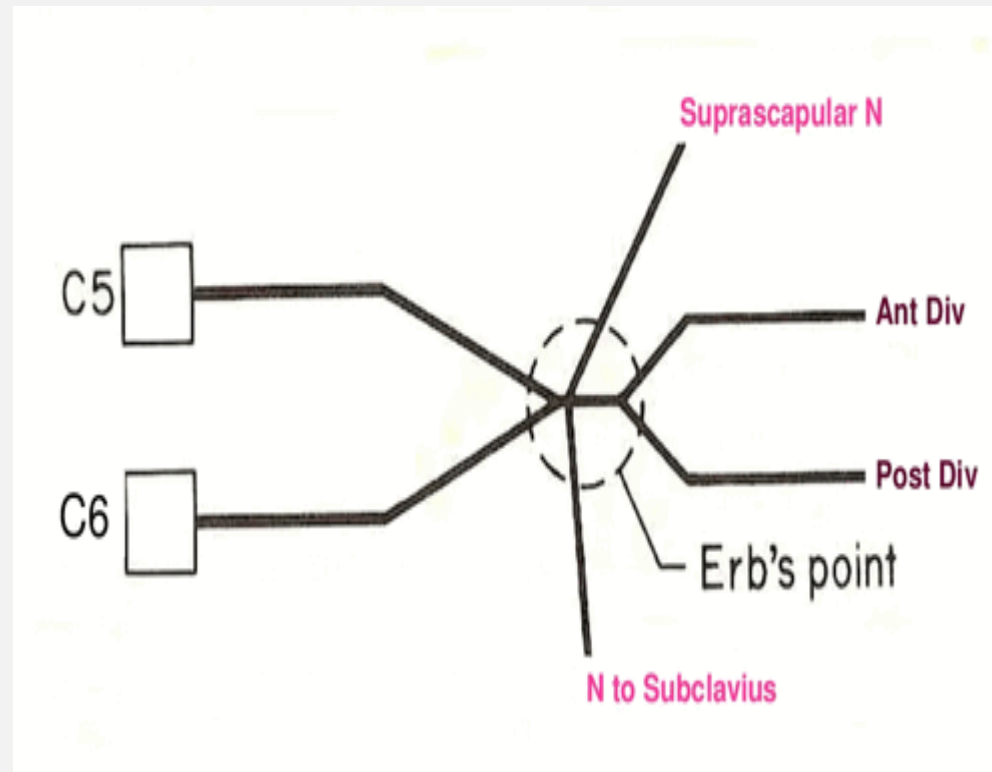
- ❑ **Upper trunk injury** (Erb's palsy):
- ❑ **Lower trunk injury** (Klumpke's palsy):



❑ Erb's point in brachial plexus?

meeting point of six nerves , which includes

- ventral rami of C5 and C6
- two branches (suprascapular and nerve to subclavius) of upper trunk
- two divisions of upper trunk



Brachial Plexus Injuries

A) Upper trunk injury (Erb's Douchenne paralysis)

- damage to the **CS and C6** fibers of the superior trunk
- **Mechanism of truma:**
stretching or tearing the upper brachial plexus in the posterior triangle region of the neck as in
 - a) Falls on the shoulder in which head & shoulder are separated
 - b) Abstracted labour
- **Nerves affected:**

Musculocutaneous, axillary, suprascapular, and N. subclavius

Motor deficits:

- 1) Paralysis of abductor of shoulder, flexor of elbow and supinator muscles leading to combined flat shoulder and **policeman tip deformity** (adduction of arm, extension & pronation of forearm).
- 2) Wasting of scapular muscles, front of arm with flat shoulder.

Sensory deficit:

- Sensory loss over lower 1/2 of deltoid, lateral aspect of arm, lateral aspect of forearm, lateral aspect of hand & lateral 2 fingers
- **Deformity:** **policeman tip deformity**



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B) Lower trunk injury (Klumpke's palsy)

damage to the C8 and T1 fibers of the lower trunk

Causes:

- occurs when upper limb is suddenly pulled upward and away from the body.
- Obstetric injury (shoulder dystocia)

Nerves affected:

C8 and T1 lesions primarily affect

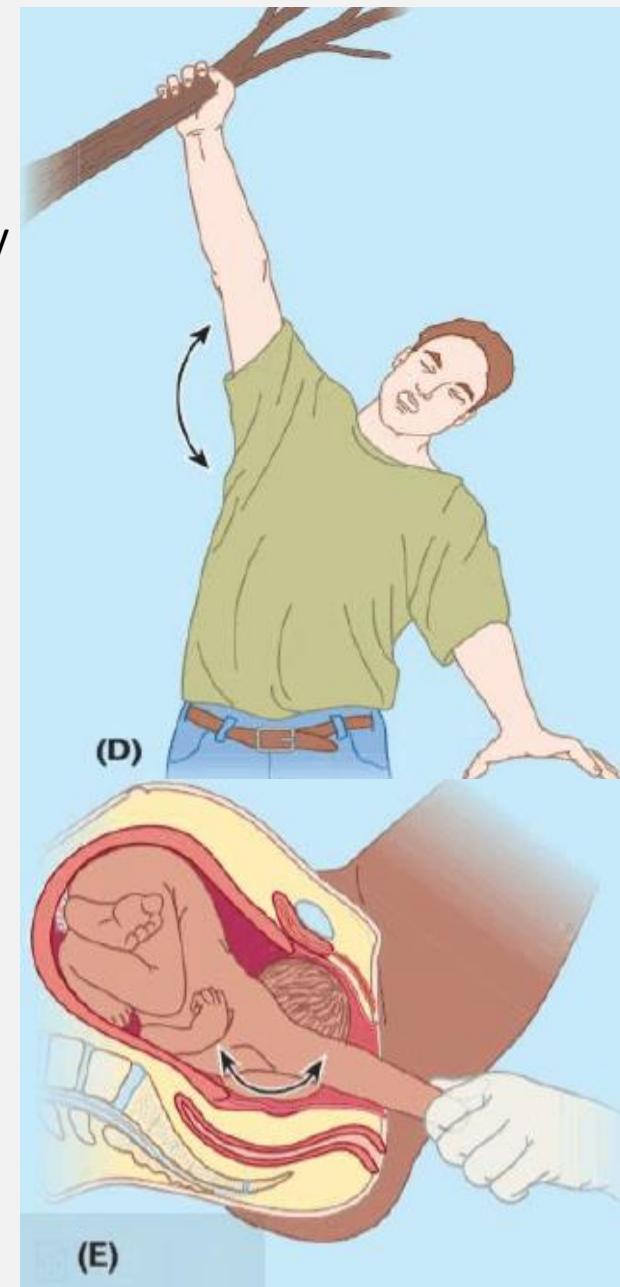
- 1) intrinsic muscles of hand innervated by ulnar nerve (**interossei & medial two lumbricals**)
- 2) some of muscles innervated by C8 and T1 fibers of median nerve (**thenar compartment**)

Motor deficits:

- 1) Patient presents with a **combined claw** hand and some degree of ape hand.
- 2) In addition, there is some weakness of the thenar muscles.

Sensory deficit:

- Sensory loss of the medial border of the forearm and the medial 1 & 1/2 digits



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Paralysis of all small muscles of hand (C8&T1)

median & Ulnar nerves

- Hyperextended MCP joints
- Flexion of IP joints

QUIZZES

Q1. What structure separates the upper trunk from the subclavian artery?

Q2. Which root contributes to ALL terminal branches of the plexus?

Q3. What finding indicates T1 root avulsion?

Critical Structures by Injury Level:

Root Avulsion (Pre- vs. Post-ganglionic):

- **Pre-ganglionic** (intraforaminal): Horner's

◦ .

Post-ganglionic (extraforaminal): **Potentially repairable.**



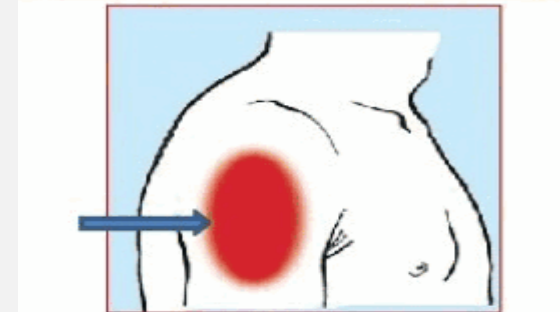
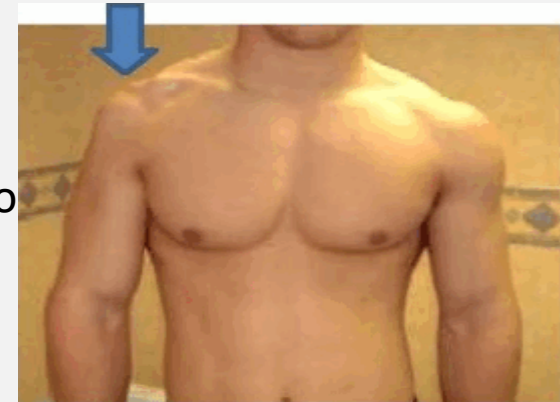
https://www.youtube.com/watch?v=MN9H_HuLBgg

Injuries Of Peripheral Nerve

1- Axillary Nerve Injury

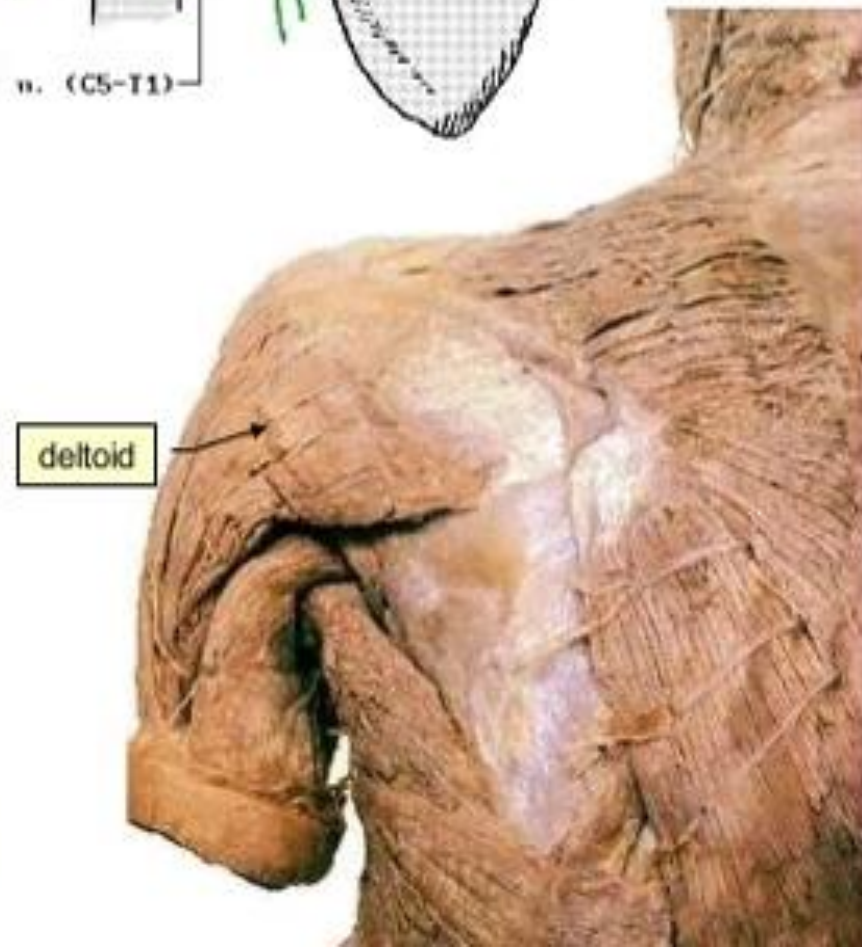
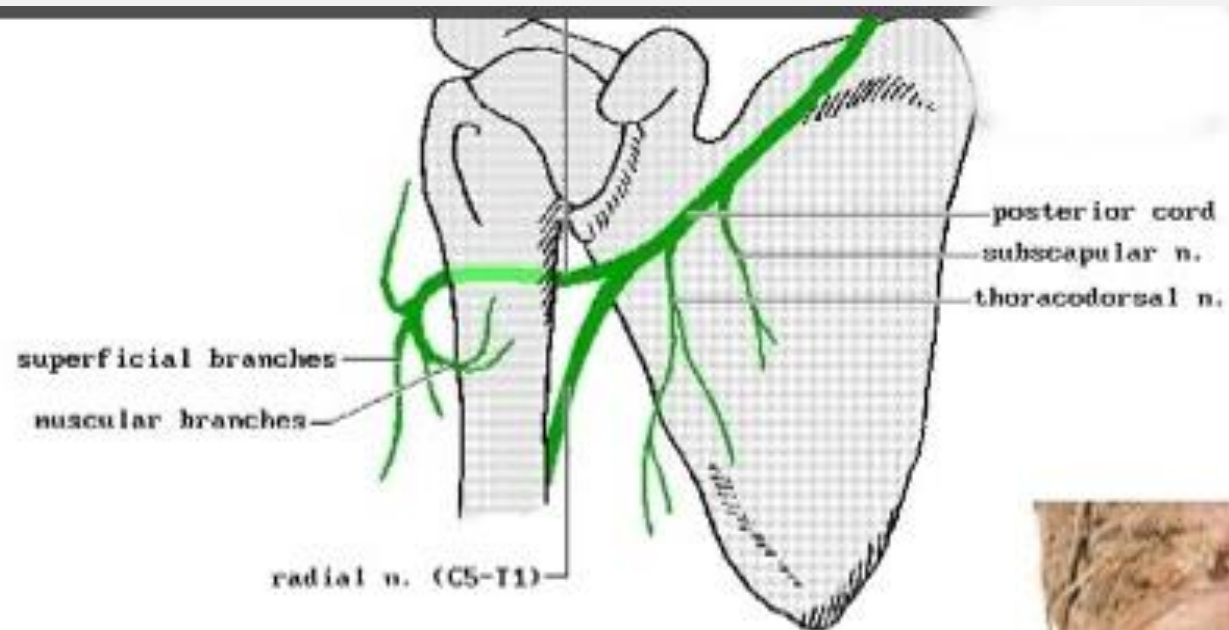
- **Root :** (C5-C6)
- **Common lesion sites:**
 - (1) fracture of the surgical neck of the humerus
 - (2) inferior dislocation of the shoulder joint.
- **Muscles innervated:** **deltoid & teres minor muscles.**
- **Motor deficits:**
 - 1) Loss of horizontal abduction (15- 110 degrees) of the arm
 - 2) Atrophy of the deltoid muscle and loss of the rounded contour of the shoulder.
 - 1) Weak lateral rotation of arm
- **Sensory deficit:**

Sensory loss of skin overlying the cap of the shoulder..



Axillary nerve

- **Origin and branches:**
 - Muscular: deltoid and teres minor.
 - Sensory: upper lateral cutaneous nerve of the arm.
- **Testing.**



2- Suprascapular nerve injury

- **Root :** (C5-C6)
- **Common lesion sites:**
Injury due to entrapment as the nerve crosses the superior border of the scapula in the scapular notch

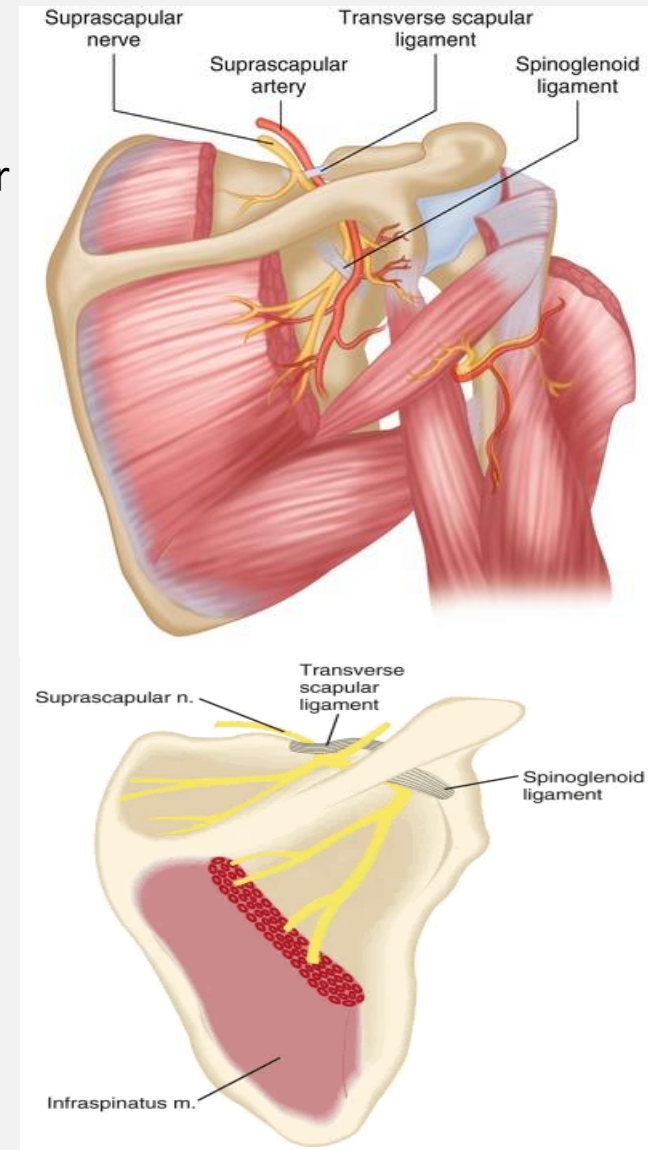
Muscles innervated:

supraspinatus & Infraspinatus.

- **Motor deficits:**
 - 1) Weakness in initiating the first 15 degrees of shoulder abduction
 - 2) Some weakness of external rotation of the shoulder.

Sensory deficit:

- No sensory loss



3- Long Thoracic Nerve injury?

- **Root :** (C5,C6 , C7)
- **Common lesion sites:**

Damaged as the nerve passes on the superficial surface of the Serratus anterior muscle by trauma to the lateral chest wall or following **mastectomy** when the nerve may be accidentally cut at the lateral aspect of the mammary gland

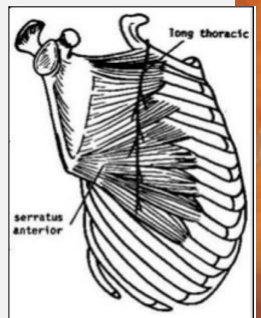
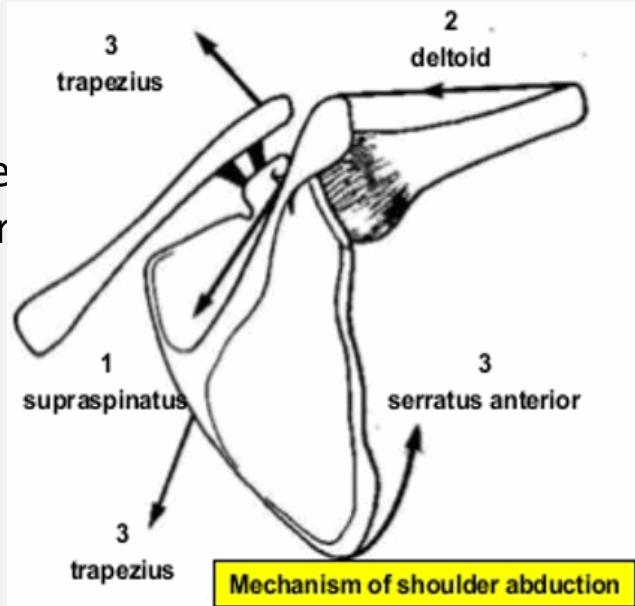
Muscles innervated:

Serratus anterior muscle .

- **Motor deficits:**
 - 1)difficulty in overhead abduction
 - 2)Weakness protraction
 - 3) Sign is a winged scapula

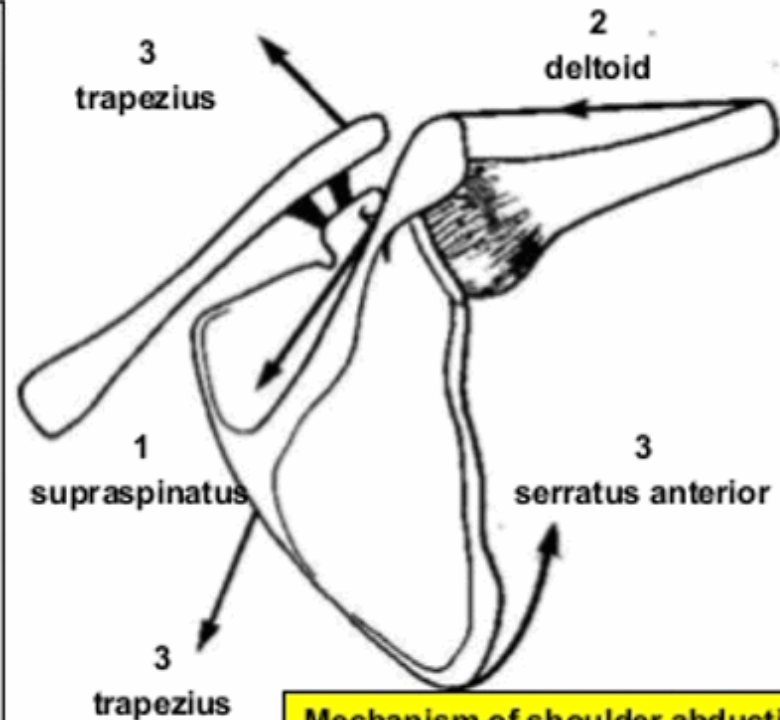
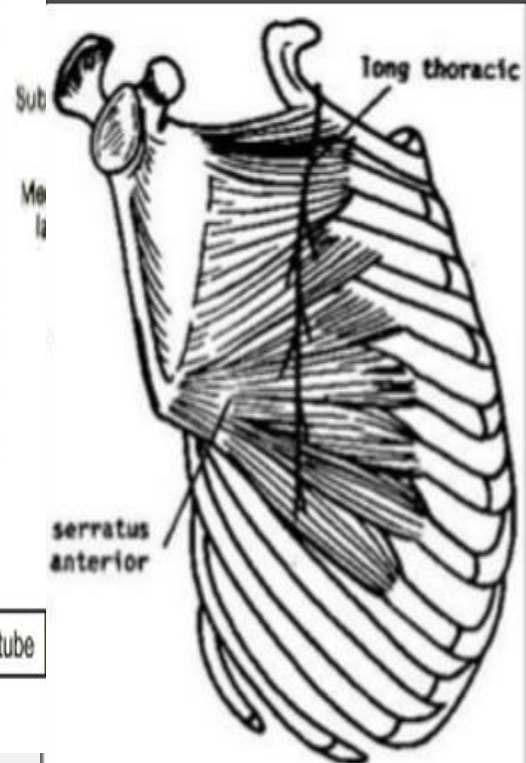
Sensory deficit:

- No sensory loss





Chest tube



Mechanism of shoulder abduction





Compete the following table with appropriate nerves suitable for the abnormalities that occur when these nerves are injured ?

Deformity	Nerve
Flatting of shoulder contour	Axillary
Wrist drop	Radial
Partial Claw hand	Ulnar
Benediction hand	Median
Winging of scapula	Long thoracic
Ape like hand	Median
Waiter's tip position	Upper trunk of brachial plexus Erb's Douchenne paralysis
Combined Claw hand	Lower trunk of brachial plexus Klumpke's palsy